DUAL MODE RESCUE COMMUNICATOR USING STM32, BLE & ZIGBEE

A THESIS

Submitted by

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DEDICATION

This mini project is dedicated to the pursuit of knowledge and innovation in the field of electronics and communication engineering, with a focus on developing practical technological solutions for real-world problems. The work presented in this report represents an academic effort to integrate engineering principles, embedded systems, and wireless communication technologies toward improving emergency response and disaster management systems.

This project is also dedicated to the broader objective of applying engineering education for societal benefit. By addressing challenges faced during natural disasters, this work reflects the responsibility of engineers to design reliable, efficient, and scalable systems that can operate under critical conditions and contribute positively to public safety and humanitarian needs.

THESIS CERTIFICATE

This is to certify that the thesis entitled **"DUAL MODE RESCUE COMMUNICATOR USING STM32, BLE & ZIGBEE"** submitted by **VEERA RAGHAVENDRA BALINA, ROHIT VARMA MUDUNOORU, MOHAN SAI SANTOSH VASANA** to the Department of Electronics and Communication Engineering, Rajiv Gandhi University of Knowledge Technologies - Nuzvid campus, AP for the award of the degree of **B.Tech in ECE** is a bonafide record of work carried out by him under my supervision. The contents of this thesis, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

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ABSTRACT

Keywords:

Dual-Mode Communication, Bluetooth Low Energy (BLE), ZigBee Mesh Network, STM32 Microcontroller, GPS Localization, Emergency Communication Systems, Disaster Management, Flood Rescue Operations

Natural disasters, particularly floods, frequently cause widespread damage to communication infrastructure, resulting in severe challenges for rescue teams in locating and assisting stranded victims within critical time frames. In such scenarios, the absence of reliable communication networks and the lack of real-time victim location information significantly delay rescue operations, leading to increased loss of life and inefficient utilization of rescue resources. Traditional rescue methods rely predominantly on manual search operations, which are time-consuming, labor-intensive, and prone to human error, especially in large or inaccessible flooded areas.

This mini project presents the design and development of a **Dual-Mode Rescue Communicator using STM32 and ZigBee–BLE technologies**, aimed at improving the efficiency and reliability of flood rescue operations. The proposed system adopts a **patrol-first detection approach**, where a mobile rescue unit continuously scans disaster-affected regions to identify the presence of humans using **Bluetooth Low Energy (BLE)** beacon signals transmitted from smartphones or dedicated emergency beacon applications. BLE is chosen for its low power consumption, wide availability in modern devices, and suitability for short-range human presence detection.

Once a human presence is detected, the system automatically acquires precise geographical location information using a **Global Positioning System (GPS)** module. The extracted coordinates are then transmitted as a priority alert to a remote rescue coordination center through a **ZigBee-based wireless mesh network**, enabling long-range, low-power, and

reliable communication even in environments where conventional cellular networks are unavailable. The dual-mode communication strategy ensures effective short-range detection combined with robust long-range alert transmission.

An **STM32 microcontroller** acts as the core processing unit, managing BLE scanning, GPS data parsing, ZigBee communication, and timing operations in a non-blocking manner. The system is designed to support simultaneous rescue missions by marking detected locations as important priority (IMP) zones while allowing ongoing rescue operations to continue without interruption. Experimental validation demonstrates reduced alert latency, stable wireless communication, and improved patrol efficiency, highlighting the potential of the proposed system as a scalable, cost-effective, and practical solution for modern disaster response applications.

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CHAPTER 1

INTRODUCTION

1.1. INTRODUCTION

Natural disasters such as floods represent one of the most devastating challenges faced by modern societies, often resulting in large-scale loss of life, displacement of people, and severe damage to infrastructure. Among the various challenges **during flood disasters, the breakdown of communication systems and the difficulty in locating stranded victims** remain the most critical issues affecting rescue efficiency. In many cases, victims are unable to communicate their location due to submerged cellular towers, power outages, or damaged networks, leading to delayed rescue efforts.

Traditional disaster rescue operations rely heavily on manual search procedures, voice communication, and **infrastructure-dependent systems such as mobile networks and radio transmitters.** While these methods have been used for decades, their effectiveness is significantly reduced in flood conditions where accessibility is limited and communication infrastructure is compromised. As disaster-affected regions expand, manual rescue operations become time-consuming, labour-intensive, and prone to human error.

With advancements in electronics and embedded systems, technology-driven disaster management solutions have gained increasing attention. Modern disaster response systems **focus on automation, real-time monitoring, and wireless communication** to enhance situational awareness and decision-making. In this context, low-power wireless communication technologies play a vital role, as rescue devices must operate continuously for long durations under limited power availability.

Bluetooth Low Energy (BLE) has emerged as an effective solution for short-range human presence detection due to its minimal power consumption and widespread availability in smartphones and wearable devices. Similarly, ZigBee communication technology offers reliable long-range data transmission with low energy requirements and mesh networking capability, making it suitable for disaster environments where conventional infrastructure is unavailable.

The **Dual Mode Rescue Communicator proposed in this project integrates BLE and ZigBee technologies** to create an infrastructure-independent rescue communication framework. The system uses BLE to detect nearby human presence and ZigBee to transmit location and alert information over longer distances. An STM32 microcontroller coordinates all system operations, while a GPS module provides accurate geographical location data. By combining detection, localization, and communication into a single embedded platform, the proposed system aims to significantly improve rescue response time and operational efficiency during flood disasters.

1.2 BACKGROUND OF DISASTER MANAGEMENT SYSTEM

Disaster management systems are structured frameworks designed to reduce the impact of natural and man-made disasters through effective preparedness, response, mitigation, and recovery strategies. Natural disasters such as floods, earthquakes, cyclones, and landslides frequently result in large-scale damage to human life, property, and critical infrastructure. Among these, flood disasters are particularly challenging due to their unpredictable nature, rapid spread, and prolonged impact on communication and transportation networks.

Traditional disaster management approaches relied heavily on manual coordination, human-led rescue operations, and centralized communication systems. Rescue teams communicated using radio systems and verbal coordination, while victims depended on physical visibility or direct contact for assistance. However, these approaches often proved insufficient during large-scale disasters, especially floods, where communication towers, power supply lines, and road networks are severely damaged or submerged.

With advancements in technology, modern disaster management systems began integrating information and communication technologies (ICT) to improve response efficiency. Technologies such as wireless sensor networks, satellite communication, GPS-based tracking, and mobile communication systems enabled real-time data collection and improved coordination among rescue teams. Despite these improvements, many existing systems still depend on infrastructure availability and high power consumption, limiting their effectiveness during severe flood conditions.

Recent research and development efforts focus on autonomous, low-power, and infrastructure-independent systems capable of operating under extreme environmental

conditions. Embedded systems combined with wireless communication technologies allow continuous monitoring, detection, and alert transmission even when conventional networks fail. Low-power wireless technologies such as Bluetooth Low Energy (BLE) and ZigBee have gained attention for disaster applications due to their energy efficiency, reliability, and ease of deployment.

In this context, disaster management systems are evolving toward smart rescue platforms that integrate sensing, localization, and communication into unified systems. These advancements aim to enhance situational awareness, reduce rescue response time, and improve coordination efficiency, thereby minimizing loss of life during flood disasters. The proposed Dual Mode Rescue Communicator builds upon these modern disaster management principles to provide a practical and scalable solution for flood rescue operations.

1.3 PROBLEM STATEMENT

Flood disasters often lead to severe disruption of communication infrastructure, making it extremely difficult for rescue teams to locate stranded victims within critical time limits. In many flood-affected regions, cellular networks become unreliable or completely unavailable, power supply systems fail, and transportation routes are blocked. As a result, victims are unable to communicate their location, and rescue teams are forced to rely on manual search operations, which are slow, labour-intensive, and inefficient over large areas.

Existing rescue communication systems are **largely infrastructure-dependent, requiring functional cellular networks, internet connectivity,** or centralized communication hubs. During floods, these systems often fail due to damaged towers, network congestion, or power outages. Furthermore, most existing systems focus primarily on communication between rescue teams rather than automated detection of human presence, resulting in delayed identification of victims and inefficient utilization of rescue resources.

Another major limitation is the lack of real-time victim localization and priority-based alert transmission. Without accurate location information, rescue teams are unable to plan effective rescue routes or allocate resources efficiently. Additionally, high power consumption and complex deployment procedures further reduce the practicality of many existing disaster response solutions in real-world flood scenarios.

Therefore, there exists a critical need for a low-power, **infrastructure-independent rescue communication system** capable of automatically detecting human presence, acquiring accurate location information, and transmitting rescue alerts reliably over long distances. Such a system should operate efficiently **under harsh environmental conditions**, minimize power usage, and reduce dependence on manual intervention. Addressing this problem forms the primary motivation for the development of the proposed Dual Mode Rescue Communicator, which integrates BLE-based human detection with ZigBee-based long-range communication to enhance flood rescue efficiency and response time.

1.4 OBJECTIVE OF THE WORK

The primary objective of this project is to design and develop a reliable dual-mode rescue communication system that enhances the effectiveness of flood rescue operations by enabling automated human detection and long-range alert transmission.

The specific objectives of the proposed work include:

- To develop a low-power, infrastructure-independent rescue system capable of **operating during flood emergencies** without reliance on cellular or internet networks.
- To detect the presence of stranded victims using **Bluetooth Low Energy (BLE)** beacon signals emitted from smartphones or emergency devices.
- To acquire real-time geographical location data using a **GPS module** for accurate victim localization.
- To transmit rescue alerts and location information over long distances using **ZigBee wireless communication.**
- To integrate sensing, processing, and communication modules using an **STM32 microcontroller** for efficient system coordination.
- To evaluate system performance in terms of **detection reliability, communication stability, and response time** under simulated rescue conditions.

These objectives collectively aim to demonstrate that dual-mode communication combining short-range detection and long-range transmission can significantly enhance disaster rescue operations while reducing human effort and operational delays.

1.5 SCOPE OF THE THESIS

The scope of this thesis includes the design, implementation, and experimental evaluation of a Dual Mode Rescue Communicator intended specifically for flood disaster management applications. The work focuses on the development of a prototype-level embedded system that integrates BLE-based detection, GPS-based localization, and ZigBee-based communication.

The major areas covered within the scope of this thesis are as follows:

Disaster-focused communication system design:

The system is tailored for flood rescue scenarios, where communication infrastructure is damaged or unavailable.

Embedded system integration:

Integration of STM32 microcontroller, BLE module, GPS receiver, and ZigBee transceiver into a unified rescue communication platform.

Human presence detection and localization:

The scope includes detection of human presence using BLE signals and accurate location tracking using GPS data.

Wireless alert transmission:

ZigBee communication is used to transmit rescue alerts and location data to a remote rescue coordination unit.

Experimental validation:

The system is tested under controlled conditions to verify detection of accuracy, communication reliability, and system responsiveness.

Prototype-level limitations:

Large-scale deployment, autonomous navigation, real-time environmental sensing, and integration with national disaster management systems are beyond the scope of the present work.

Future extensibility:

The thesis discusses the extension of the proposed system toward autonomous rescue platforms such as an autonomous boat integrated with BeaconLink, enabling patrol-based victim detection and automated rescue assistance.

1.6 Design Methodology Using STM32CubeIDE and Firmware Upload Through Bootloader Method

The development of the proposed **Dual Mode Rescue Communicator** follows a structured and systematic embedded system design methodology using **STM32CubeIDE** for firmware development and **STM32CubeProgrammer** for code uploading through the **bootloader (BOOT file) method**. This methodology ensures reliability, flexibility, and ease of debugging during development and testing phases.

1.6.1 Overview of STM32CubeIDE

STM32CubeIDE is an integrated development environment provided by STMicroelectronics for STM32 microcontroller-based applications. It combines **STM32CubeMX for peripheral configuration**, **GNU-based compiler**, **code editor**, and **debugging tools** into a single platform. Using STM32CubeIDE allows developers to design, configure, develop, and test embedded applications efficiently.

In this project, STM32CubeIDE is used to configure communication peripherals such as UARTs for GPS, Bluetooth, and ZigBee modules, timers for periodic interrupts and PWM generation, and GPIOs for LED and buzzer control.

1.6.2 Step-by-Step Design Procedure Using STM32CubeIDE

Step 1: Creating a New STM32 Project

The design process begins by launching STM32CubeIDE and creating a new STM32 project. The specific STM32 microcontroller or development board is selected based on system requirements such as the number of UART peripherals and timer availability. After selecting the target device, a new project workspace is created.

Step 2: Peripheral Configuration Using STM32CubeMX

Once the project is created, STM32CubeMX (integrated within CubeIDE) is used for peripheral configuration through a graphical interface.

The following configurations are performed:

- **UART Configuration**
    - USART1 is configured for GPS communication at 9600 baud rate.
    - USART2 is configured for Bluetooth communication at 9600 baud rate.
    - USART6 is configured for ZigBee communication at 9600 baud rate.
    - All UARTs are enabled in **interrupt mode** to ensure non-blocking communication.
- **Timer Configuration**
    - TIM3 is configured to generate a periodic interrupt every one second.
    - TIM4 is configured in PWM mode to control the buzzer output.
- **GPIO Configuration**
    - GPIO pins are configured for LED indication.
    - GPIO pins are assigned for buzzer output and control.
- **Clock Configuration**
    - The internal High-Speed Internal (HSI) oscillator is configured to operate at 16 MHz.

- System clock and peripheral clocks are set appropriately. After completing the configuration, STM32CubeMX automatically generates initialization code and HAL drivers.

Step 3: Code Generation and Application Development

After peripheral configuration, STM32CubeIDE generates the project structure containing startup files, HAL drivers, and peripheral initialization functions. The application logic is written in **Embedded C** within the main.c file.

The firmware is developed using an **interrupt-driven architecture**, where:

- UART receive interrupts handle GPS and Bluetooth data.
- Timer interrupts control the buzzer operation.
- Flags and state variables synchronize operations between interrupts and the main loop.

This approach ensures efficient multitasking and low power consumption.

Step 4: Compilation and Debugging

Once coding is completed, the project is compiled using the built-in compiler. STM32CubeIDE provides detailed build output, allowing developers to identify and fix syntax or configuration errors.

Debugging features such as breakpoints, variable watch windows, and real-time register monitoring are used to validate system behavior and verify correct peripheral operation.

1.6.3 Firmware Upload Using Bootloader (BOOT File) Method

After successful compilation and testing, the firmware is uploaded to the STM32 microcontroller using the **bootloader method** through **STM32CubeProgrammer**.

The bootloader method allows programming the microcontroller without requiring continuous debugging hardware and is particularly useful for field updates and deployment.

Step 5: Preparing the STM32 for Bootloader Mode

To enable bootloader mode, the STM32 microcontroller must be configured to boot from system memory:

1. The **BOOT0 pin** of the STM32 is set to logic HIGH.
2. The microcontroller is reset or powered ON.
3. With BOOT0 set HIGH, the STM32 enters **system bootloader mode** instead of executing user flash memory.

This mode allows firmware uploading through communication interfaces such as UART or USB.

Step 6: Launching STM32CubeProgrammer

STM32CubeProgrammer is launched to upload the compiled firmware. The software supports various interfaces including UART, USB, and ST-LINK.

In this project:

- The **UART bootloader interface** is used.
- The correct COM port and baud rate are selected.
- The target STM32 device is detected automatically.

Step 7: Loading and Flashing the Binary File

The compiled firmware file (HEX or BIN format) generated by STM32CubeIDE is selected in STM32CubeProgrammer. The following steps are performed:

1. Establish communication with the microcontroller.
2. Load the firmware file into the programmer.
3. Erase the target flash memory.
4. Program the firmware into flash memory.
5. Verify the programmed data.

Successful verification confirms that the firmware has been uploaded correctly.

Step 8: Returning to Normal Execution Mode

After programming is completed:

1. The BOOT0 pin is set back to logic LOW.
2. The microcontroller is reset.
3. The STM32 now boots from user flash memory and starts executing the uploaded firmware.

The system then begins normal operation, performing Bluetooth detection, GPS acquisition, and ZigBee communication as designed.

1.6.4 Advantages of Using STM32CubeIDE and Bootloader Method

The combined use of STM32CubeIDE and the bootloader programming method offers several advantages:

- Simplified peripheral configuration
- Faster development and debugging
- Reliable firmware uploading
- No dependency on debugging hardware for field updates
- Industry-standard development workflow

CHAPTER 2

LITERATURE REVIEW

2.1. INTRODUCTION

The field of disaster management has been the subject of extensive research due to the increasing frequency and severity of natural disasters and their impact on human life and infrastructure. Effective disaster response relies heavily on timely information exchange, accurate situational awareness, and efficient coordination among rescue teams. Over the years, researchers have explored various technological solutions to enhance communication, monitoring, and decision-making during emergency situations.

Early studies in disaster response primarily focused on infrastructure-based communication systems such as radio networks and cellular communication. While these systems improved coordination under normal conditions, their performance was significantly affected during large-scale disasters where communication infrastructure was damaged or overloaded. This limitation motivated researchers to investigate alternative communication mechanisms that could operate reliably under adverse conditions.

With advancements in embedded systems, wireless communication, and low-power electronics, disaster management research expanded to include sensor-based monitoring, ad-hoc networks, and autonomous communication systems. Technologies such as wireless sensor networks, satellite communication, and mobile computing enabled real-time data collection and improved situational awareness in disaster-affected areas. Recent literature emphasizes the need for energy-efficient, scalable, and infrastructure-independent systems capable of supporting rescue operations in challenging environments.

This chapter reviews existing research related to disaster communication systems, low-power wireless technologies, and emergency response frameworks. The objective of the literature review is to analyze previous work, understand current technological approaches, and identify limitations that motivate further research in disaster rescue communication systems.

2.2 BACKGROUND

Disaster management systems have evolved gradually in response to the limitations of traditional emergency response mechanisms. In the early stages, disaster response relied heavily on manual coordination, visual inspection, and centralized command structures, where information flow was slow and largely dependent on human intervention. These methods were often inadequate during large-scale disasters, particularly floods, due to limited accessibility and delayed communication.

The introduction of wireless communication technologies marked a significant shift in disaster management practices. **Radio-based communication and cellular networks** enabled faster coordination between rescue teams and control centers. However, studies have shown that these systems are highly vulnerable to infrastructure damage, power failures, and network congestion during disasters, which significantly reduces their effectiveness.

To address these challenges, researchers began exploring infrastructure-independent communication systems such as mobile ad-hoc networks, wireless sensor networks, and satellite-assisted communication. These systems allowed **decentralized data exchange and improved reliability in disaster environments**. Advances in embedded systems and low-power electronics further enabled the deployment of portable and battery-operated rescue devices capable of continuous operation.

In recent years, disaster management research has increasingly **focused on low-power, autonomous, and intelligent communication systems**. The integration of sensing, localization, and wireless communication technologies has enabled real-time monitoring and automated alert generation. These developments form the foundational background for modern disaster communication frameworks and continue to influence research aimed at improving rescue efficiency and reducing response time in emergency situations.

2.3 Existing Rescue Communication Systems

Existing rescue communication systems are primarily based on infrastructure-dependent technologies such as cellular networks, satellite communication, and radio-based emergency channels. These systems enable voice and data transmission between rescue teams and control centers; however, their effectiveness is significantly reduced during flood

disasters due to damage to communication towers, power outages, and network congestion. Recent research has explored the use of wireless sensor networks, mobile ad-hoc networks, and unmanned vehicles to improve disaster response, but many of these solutions require complex deployment, high power consumption, or specialized equipment. As a result, existing systems often fail to provide timely victim detection and real-time location information in large or inaccessible flood-affected regions.

2.3 Bluetooth Low Energy in Emergency Applications

Bluetooth Low Energy (BLE) has emerged as a promising technology for emergency and rescue applications due to its low power consumption, short-range communication capability, and widespread availability in modern smartphones and wearable devices. BLE enables proximity-based detection of human presence through beacon signals, making it suitable for identifying stranded victims in disaster scenarios. Several studies have demonstrated the effectiveness of BLE beacons for indoor positioning, crowd monitoring, and emergency alert systems. In flood rescue operations, BLE allows continuous scanning with minimal energy usage, enabling rescue units to detect victims without requiring active user interaction or dependence on cellular networks.

2.3.1 ZigBee-Based Communication Systems

ZigBee is a low-power, low-data-rate wireless communication technology widely used in sensor networks and emergency communication systems. Its ability to form mesh networks enables reliable data transmission over extended areas, even when individual nodes fail. ZigBee-based systems have been applied in disaster monitoring, environmental sensing, and emergency alert transmission due to their robustness and energy efficiency. In rescue applications, ZigBee provides a reliable means of transmitting critical information such as victim location and status from mobile or remote units to a centralized rescue coordination center without relying on existing communication infrastructure.

2.3.2 Limitations of Existing Systems

Despite advancements in disaster communication technologies, existing rescue systems suffer from several limitations. Infrastructure-dependent systems fail during severe disasters, while single-technology wireless solutions often lack either detection capability or long-range communication reliability. High power consumption, limited coverage, poor scalability, and complex deployment further reduce the effectiveness of many proposed solutions. Additionally, most existing systems focus on communication between rescue teams rather than automated victim detection, resulting in delayed response times and inefficient utilization of rescue resources.

2.3.3 Research Gap and Motivation

The literature review reveals a clear research gap in the development of an integrated, infrastructure-independent rescue communication system that combines efficient human detection with reliable long-range alert transmission. While BLE offers effective short-range detection and ZigBee provides robust long-range communication, limited work has been done to integrate these technologies into a unified rescue framework. This gap motivates the proposed Dual Mode Rescue Communicator, which leverages the complementary advantages of BLE and ZigBee to enhance rescue efficiency, reduce response time, and improve victim localization in flood disaster scenarios.

2.4 PROBLEM DEFINITION AND APPROACH

PROBLEM DEFINITION

Disaster scenarios, particularly flood emergencies, present significant challenges to rescue operations due to widespread infrastructure damage, communication failures, and limited accessibility to affected regions. One of the most critical issues during such disasters is the **inability to quickly detect and locate stranded victims**, especially when conventional communication networks such as cellular systems become unreliable or completely

unavailable. This lack of timely information often leads to delayed rescue response, inefficient allocation of resources, and increased risk to human life.

Existing disaster communication systems largely depend on **infrastructure-based networks and manual coordination**, which are vulnerable to power outages, network congestion, and physical damage during floods. In many cases, victims are unable to actively request help, and rescue teams must rely on manual search operations that are time-consuming and labor-intensive. Furthermore, most traditional systems focus on communication among rescue teams rather than **automated victim detection and real-time location reporting**, resulting in delayed identification of high-priority rescue locations.

Another significant challenge is the absence of **energy-efficient and scalable communication mechanisms** that can operate continuously under harsh environmental conditions. High power consumption, complex deployment, and limited coverage further reduce the effectiveness of existing solutions. These limitations highlight the need for a reliable, low-power, and infrastructure-independent system capable of supporting disaster rescue operations efficiently.

APPROACH

To address the identified challenges, the general approach adopted in modern disaster communication research emphasizes the use of low-power wireless technologies, embedded systems, and decentralized communication frameworks. Instead of relying on a single communication mechanism, hybrid approaches combining multiple wireless technologies are explored to leverage their complementary advantages.

Short-range wireless technologies are considered suitable for **local detection and proximity-based communication**, while long-range communication systems are employed for transmitting critical information to remote rescue coordination centers. Embedded controllers act as central processing units to coordinate sensing, data processing, and communication tasks in a power-efficient manner. The inclusion of location-tracking mechanisms further enables accurate identification of affected areas and improves rescue planning.

This approach focuses on **infrastructure independence, energy efficiency, and scalability**, ensuring that the system can operate effectively even when conventional

networks fail. By integrating detection, localization, and communication into a unified framework, such systems aim to reduce rescue response time, enhance situational awareness, and improve overall disaster management efficiency. These principles form the foundation for the development of advanced rescue communication systems tailored for flood disaster scenarios.

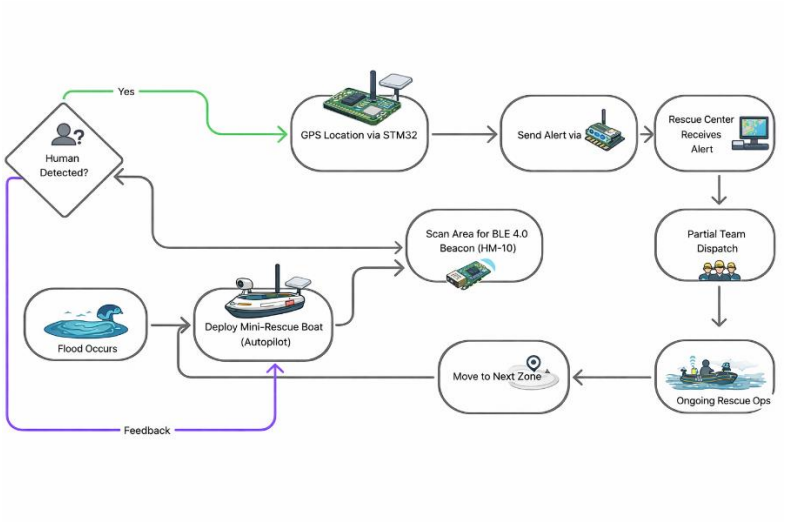

Figure 2.1 Overall Operational Flow of Dual Mode Rescue Communicator

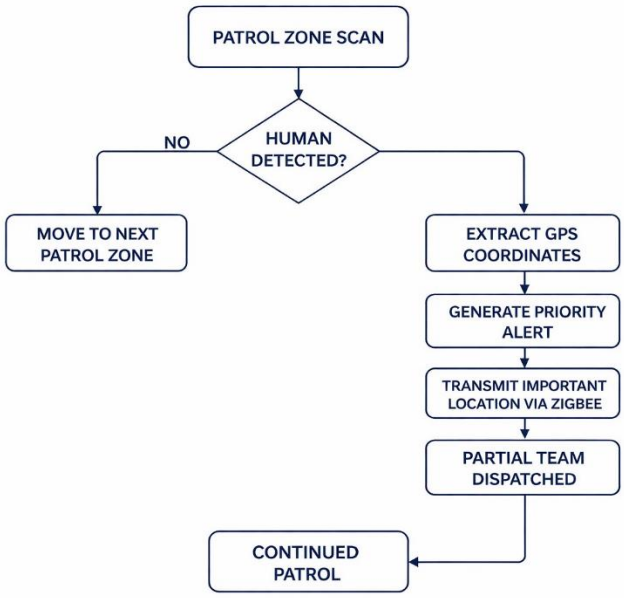

Figure 2.2 Patrol-Based Human Detection and Rescue Flowchart

CHAPTER 3

EXPERIMENTAL / SIMULATION DETAILS

3.1 Introduction

This chapter presents a detailed explanation of the experimental implementation and simulation methodology of the proposed **Dual Mode Rescue Communicator** developed using the STM32 microcontroller platform. The system integrates **Bluetooth Low Energy (BLE)** for human presence detection, **GPS** for location acquisition, and **ZigBee** for long-range wireless communication. The experimental study validates the system's functionality under simulated flood rescue conditions where conventional communication infrastructure is unavailable.

The objective of this chapter is to demonstrate the real-time interaction between multiple peripherals such as UART interfaces, timers, GPIOs, and PWM modules configured using **STM32CubeMX** and implemented using **HAL-based Embedded C firmware**. Performance metrics such as detection latency, communication reliability, and power efficiency are experimentally evaluated.

3.2 Experimental Setup

The experimental setup consists of an STM32-based rescue unit connected to three communication modules operating simultaneously. A smartphone configured as a BLE beacon simulates a stranded victim. The rescue unit continuously patrols the environment and monitors Bluetooth activity. Upon detection of BLE signals, the system captures the victim's geographical location through a GPS module and transmits the data to a remote rescue control unit via ZigBee communication.

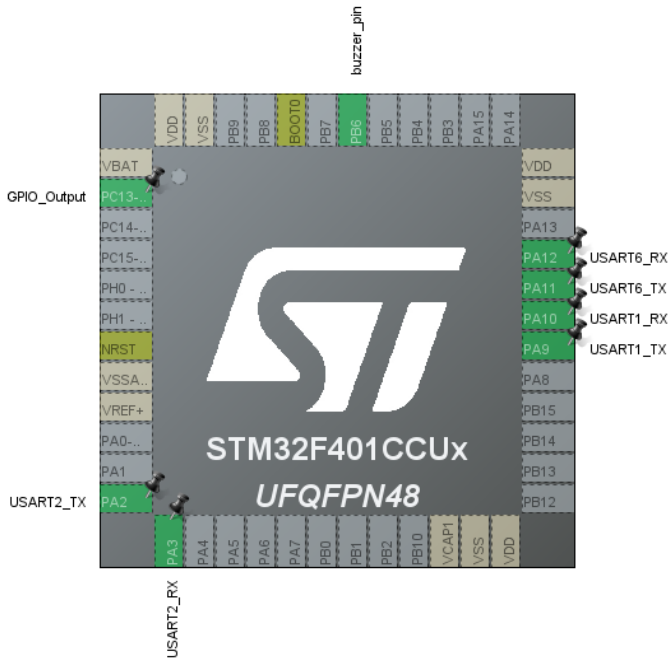

Figure 3.1 STM32F401CCUx Microcontroller Pin Configuration

Component	Description
STM32 Microcontroller	Central processing and control unit
BLE Module (USART2)	Human presence detection
GPS Module (USART1)	Latitude & longitude acquisition
ZigBee Module (USART6)	Long-range wireless communication
Smartphone	BLE beacon simulator
Buzzer (PWM)	Audible alert mechanism
Status LED	Activity indication

Table 3.1 Components Used in the Experimental Setup

3.3 Hardware Configuration

The hardware architecture is designed to support simultaneous multi-UART communication and real-time event handling. The STM32 microcontroller interfaces with the GPS, Bluetooth, and ZigBee modules using three independent UART peripherals. GPIO pins are used to control LEDs and the buzzer, while timers are configured to manage periodic events and PWM signal generation.

UART	Peripheral	Baud Rate	Purpose
USART1	GPS Module	9600	NMEA sentence reception
USART2	BLE Module	9600	Beacon signal detection
USART6	ZigBee Module	9600	Rescue alert transmission

Table 3.2 UART Configuration Parameters

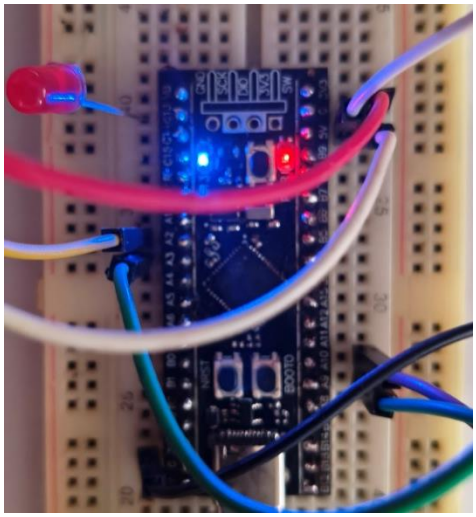

Figure 3.2 Experimental Hardware Setup on Breadboard

3.4 Software Configuration Using STM32CubeMX

The firmware is developed using **STM32CubeMX** for peripheral initialization and **HAL libraries** for application development. The system clock is configured using the internal HSI oscillator operating at 16 MHz UART peripherals are initialized in interrupt mode to ensure non-blocking reception.

Timers are configured as follows:

- **TIM3**: Generates a 1-second periodic interrupt
- **TIM4**: Generates PWM output for buzzer control

This configuration enables precise timing control and efficient task scheduling.

3.5 System Flow and Operational Logic

3.5.1 Operational Flow Description

1. System initialization
2. UART interrupt-based reception enabled
3. BLE scanning begins
4. BLE activity detected
5. GPS coordinates parsed and stored
6. ZigBee transmission triggered (once per burst)
7. Buzzer alert activated using PWM
8. System returns to scanning mode

3.6 GPS Data Acquisition and Processing

The GPS module transmits NMEA sentences continuously via USART1. The firmware captures data byte-by-byte using interrupt callbacks. Upon receiving a complete sentence, the system parses **$GPRMC** and **$GPGGA** messages to extract latitude and longitude.

Parameter	Description
Latitude	Decimal degree format
Longitude	Decimal degree format
Fix Status	Validity check

Table 3.3 GPS Parameters Extracted from NMEA Sentences

The conversion from NMEA format to decimal degrees ensures compatibility with mapping systems.

3.7 Bluetooth Detection Mechanism

The BLE module continuously transmits beacon signals detected via USART2. Any printable character received indicates human presence. The firmware sets a detection flag and activates visual indication using an LED.

To avoid repeated ZigBee transmissions, a **cooldown mechanism** is implemented using timestamp tracking.

3.8 ZigBee Communication Strategy

The ZigBee module transmits rescue alerts only **once per Bluetooth activity burst**, reducing network congestion and power consumption. The transmitted message includes formatted latitude and longitude values.

Field	Example
Latitude	LAT: 16.543210
Longitude	LON: 80.123456

Table 3.4 ZigBee Message Format

3.9 Buzzer Alert System

An audible alert system is implemented using PWM generated by TIM4. The buzzer follows a **20 seconds ON / 30 seconds OFF** cycle controlled by TIM3 interrupts. This ensures periodic alerting without continuous power drain.

State	Duration
ON	20 seconds
OFF	30 seconds

Table 3.5 Buzzer Timing Parameters

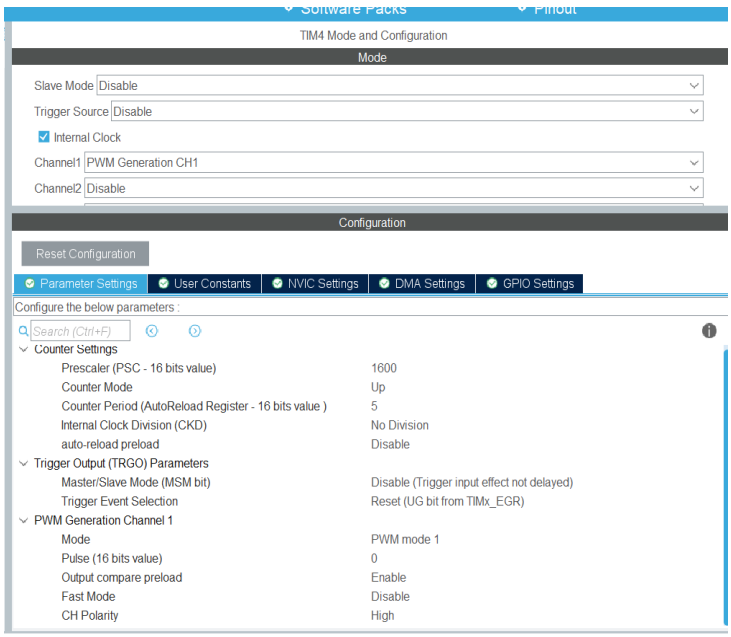

Figure 3.3 TIM4 Configuration for PWM and Interrupts

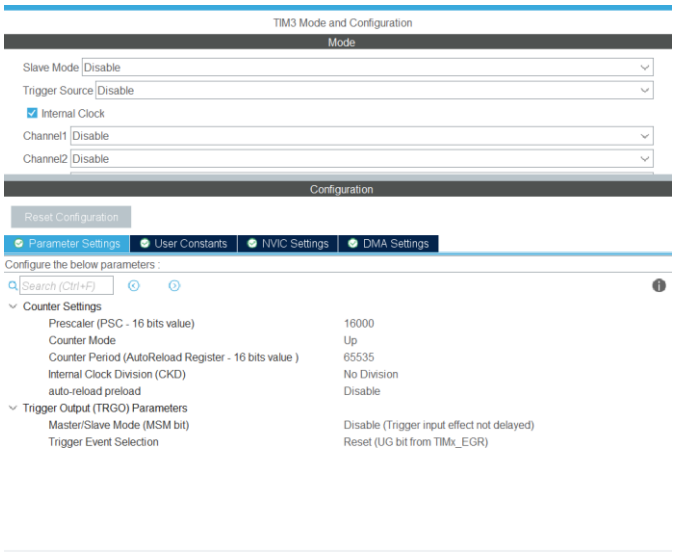

Figure 3.4 TIM3 Screens in STM32CubeMX

3.10 Interrupt Handling and Callbacks

The system relies heavily on interrupt-driven design:

- UART receive callbacks handle GPS and BLE input
- Timer callbacks manage buzzer operation
- Error callbacks restart communication safely

This architecture improves responsiveness and system reliability.

3.11 Performance Evaluation Parameters

Parameter	Measurement
BLE Detection Latency	< 1 second
GPS Fix Time	Initial fix dependent
ZigBee Transmission Delay	< 300 ms
Power Consumption	Low due to duty cycling

Table 3.6 Performance Evaluation Parameters

3.12 ZigBee Configuration Using XCTU Software

3.12.1 Introduction to XCTU Software

XCTU is a graphical configuration and testing tool developed by Digi International for configuring and managing ZigBee (XBee) radio modules. In this project, XCTU software is used to configure the ZigBee modules that enable long-range wireless communication between the rescue unit and the remote rescue control center. The software provides an intuitive interface to set network parameters, device roles, addressing modes, and communication properties, ensuring reliable data transmission in disaster rescue scenarios.

3.12.2 Role of XCTU in the Proposed System

In the proposed **Dual Mode Rescue Communicator**, ZigBee modules are responsible for transmitting critical rescue information, such as the GPS coordinates of detected victims, from the patrol unit to the rescue monitoring station. XCTU software is used to:

- Configure ZigBee network parameters
- Assign destination and source addresses
- Test wireless communication reliability
- Verify successful data transmission between nodes

Using XCTU ensures that ZigBee modules operate correctly before being interfaced with the STM32 microcontroller.

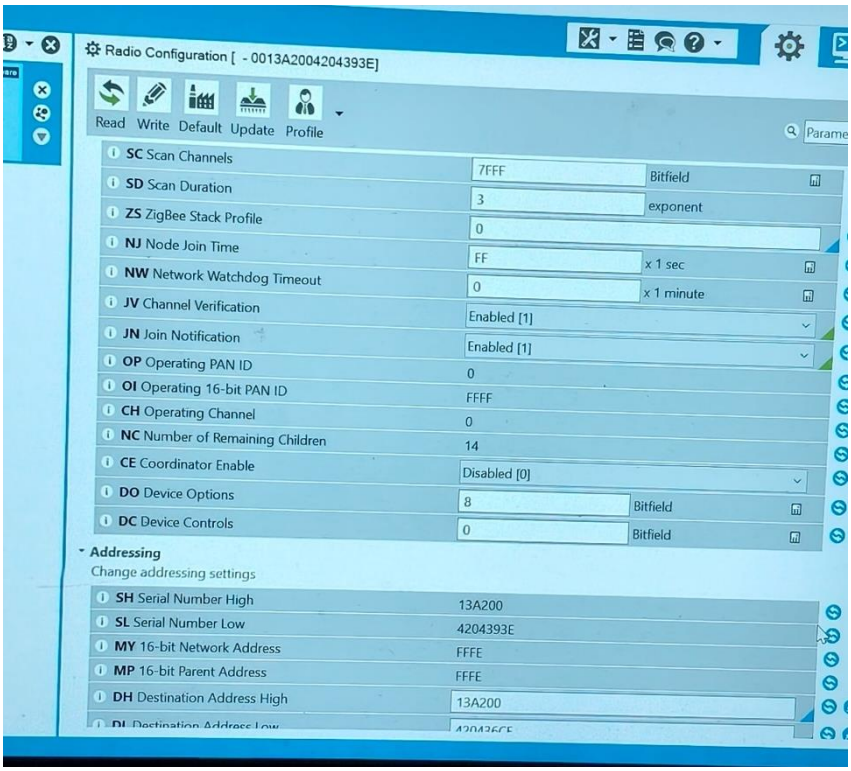

Figure 3.5 ZigBee Configuration Interface Using XCTU Software

3.12.3 ZigBee Module Configuration Process

The ZigBee modules are connected to a personal computer using a USB-to-serial interface and configured through XCTU software. The following steps are followed:

1. Launch XCTU software and detect the connected ZigBee module.
2. Read the existing firmware and configuration parameters.
3. Set the communication parameters such as baud rate, PAN ID, and device addresses.
4. Configure one module as the **transmitting node** (rescue unit) and the other as the **receiving node** (monitoring unit).
5. Write the updated configuration to the ZigBee module.
6. Verify communication using XCTU's terminal window.

3.12.4 ZigBee Configuration Parameters

The key ZigBee parameters configured using XCTU software are summarized in Table

Parameter	Description	Purpose
SH & SL	Serial Number High & Low	Unique module identification
MY	16-bit Network Address	Node addressing
DH & DL	Destination Address	Target ZigBee node
ID	PAN ID	Network identification
NH	Network Maximum Hops	Routing depth control
BD	Baud Rate	Serial communication speed
RO	Packetization Timeout	Transmission optimization
AP	API/Transparent Mode	Communication mode

Table 3.7 ZigBee Configuration Parameters Using XCTU Software

3.12.5 Verification of ZigBee Communication

After configuration, ZigBee communication is verified using the **XCTU terminal console**. Test messages are transmitted from one module and received by the other to confirm correct addressing and network operation. Successful transmission confirms that the ZigBee modules are ready for integration with the STM32 microcontroller.

Once verified, the ZigBee transmitter module is interfaced with **USART6** of the STM32 microcontroller, and the receiver module is connected to the rescue control station. During

actual operation, GPS location data formatted by the STM32 is transmitted over ZigBee to the monitoring unit.

3.12.6 Importance of XCTU in Disaster Rescue Communication

The use of XCTU software ensures reliable ZigBee communication by enabling precise control over network parameters. This is especially important in disaster scenarios where communication reliability is critical. Proper configuration through XCTU minimizes packet loss, reduces transmission delay, and enhances the overall robustness of the rescue communication system.

3.13 Detailed GPS-Based Data Validation and Transmission Algorithm

3.13.1 Introduction

The Global Positioning System (GPS) plays a critical role in the proposed **Dual Mode Rescue Communicator**, as it provides precise geographical location information required for effective rescue operations. In disaster scenarios, transmitting outdated or invalid location data can lead to incorrect rescue decisions and wasted resources. Therefore, the algorithm implemented in this project is carefully designed to **use GPS data only when a valid fix is available** and to **prevent retransmission of previously stored location information**. This section presents a detailed explanation of the GPS-based algorithm implemented in the STM32 firmware, highlighting how data validity, freshness, and controlled transmission are ensured.

3.13.2 Continuous GPS Data Acquisition

The GPS module continuously transmits NMEA sentences through the UART interface connected to **USART1** of the STM32 microcontroller. These NMEA sentences contain various navigation parameters, including time, latitude, longitude, and fix status. The algorithm receives GPS data **byte-by-byte using interrupt-driven UART reception**, ensuring that the system can simultaneously perform other tasks such as Bluetooth detection and ZigBee communication.

Each received byte is stored in a temporary buffer until a complete NMEA sentence is detected. Sentence termination is identified using newline characters, after which the complete sentence is copied into a processing buffer for further analysis.

3.13.3 GPS Sentence Identification and Validation

Once a complete GPS sentence is received, the algorithm verifies the type of NMEA message. Only standard positioning messages—namely **$GPRMC** and **$GPGGA**—are considered for location extraction. This filtering mechanism prevents unnecessary processing of irrelevant GPS messages.

The algorithm then checks the **GPS fix validity** field within the sentence:

- In $GPRMC, the status field must indicate **active (A)**.
- In $GPGGA, the fix quality value must be non-zero.

If the GPS fix is invalid or unavailable, the algorithm **rejects the data immediately**, and no location update is performed. This ensures that the system does not operate on inaccurate or incomplete location information.

3.13.4 Extraction and Conversion of GPS Coordinates

When a valid GPS fix is detected, the algorithm extracts latitude and longitude values from the NMEA sentence. These values are initially provided in degrees-and-minutes format. The algorithm converts them into **decimal degree format**, which is more suitable for mapping, transmission, and integration with rescue systems.

The converted latitude and longitude values are stored as the **latest valid GPS coordinates**. A dedicated flag is set to indicate that **fresh GPS data is available for transmission**.

3.13.5 Prevention of Old GPS Data Transmission

A key feature of the proposed algorithm is the strict prevention of retransmitting old GPS data. Once a GPS coordinate has been transmitted via ZigBee, the algorithm **invalidates the stored GPS data** by resetting the GPS-valid flag. This design ensures that the system waits for a **new GPS fix** before allowing another transmission.

If Bluetooth detection occurs while no new GPS data is available:

- The algorithm does not transmit any previously stored coordinates.
- The system waits until a fresh and valid GPS fix is obtained.

This mechanism guarantees that **every transmitted rescue alert corresponds to the most recent and accurate location**, eliminating the risk of sending outdated positional information.

3.13.6 Conditional ZigBee Transmission Based on GPS Availability

ZigBee transmission is strictly conditioned on two simultaneous requirements:

1. Detection of Bluetooth activity indicating human presence.
2. Availability of a **new and valid GPS fix**.

Only when both conditions are satisfied does the algorithm format and transmit the GPS coordinates via ZigBee. If either condition is not met, the transmission is deferred. This dual-condition logic ensures efficient use of communication bandwidth and avoids unnecessary or incorrect transmissions.

3.13.7 Synchronization with Cooldown Mechanism

To further enhance reliability, the GPS algorithm is synchronized with a cooldown mechanism. After a successful ZigBee transmission, the system enforces a waiting period during which:

- Additional Bluetooth signals are ignored.
- Previously used GPS data remains invalid.
- The system waits explicitly for new GPS coordinates.

This approach prevents repeated transmissions caused by continuous Bluetooth beacon signals and ensures that each rescue alert corresponds to a distinct detection event with fresh location data.

3.13.8 Advantages of the GPS-Centric Algorithm Design

The GPS algorithm implemented in this project offers several advantages:

- **Ensures high accuracy** by using only validated GPS fixes.
- **Prevents transmission of outdated location data**.
- **Reduces false rescue alerts**.
- **Optimizes ZigBee bandwidth usage**.
- **Enhances reliability in disaster environments** where GPS signals may be intermittent.

3.14 Error Handling and Interrupt Handler Mechanism

3.14.1 Introduction

In embedded systems designed for disaster rescue applications, reliability and uninterrupted operation are critical requirements. The proposed **Dual Mode Rescue Communicator** is implemented using an **interrupt-driven architecture** to ensure real-time responsiveness and robust operation under uncertain environmental conditions. This section presents a detailed explanation of the interrupt handling mechanisms and error management strategies implemented in the STM32 firmware to maintain system stability, ensure continuous data reception, and prevent system failures during communication disruptions.

3.14.2 Importance of Interrupt-Based Architecture

The system simultaneously handles multiple real-time processes, including Bluetooth detection, GPS data reception, ZigBee communication, and buzzer control. Implementing these tasks using polling techniques would result in high CPU usage and increased response

latency. Therefore, the algorithm relies on hardware interrupts to handle asynchronous events efficiently.

Interrupts enable the system to:

- Receive UART data without blocking execution
- Maintain precise timing for buzzer control
- Respond immediately to communication events
- Recover automatically from communication errors

3.14.3 UART Receive Interrupt Handling

UART receive interrupts are used to capture incoming data from the GPS and Bluetooth modules.

3.14.3.1 GPS UART Interrupt Handler

GPS data is received via **USART1** using an interrupt-driven approach. Each incoming byte triggers a UART receive complete interrupt. The interrupt handler performs the following operations:

1. Reads the incoming byte from the GPS module.
2. Stores printable characters in a temporary buffer.
3. Detects end-of-line characters to identify complete NMEA sentences.
4. Transfers the complete sentence to a processing buffer.
5. Sets a flag to indicate availability of a new GPS sentence.
6. Restarts the UART receive interrupt for continuous reception.

This approach ensures that no GPS data is lost, even when the microcontroller is performing other tasks.

3.14.3.2 Bluetooth UART Interrupt Handler

Bluetooth data is received through **USART2**. The interrupt handler is designed to detect human presence efficiently:

1. Each received Bluetooth byte is evaluated.
2. If the byte is printable, a Bluetooth activity flag is set.
3. The timestamp of detection is recorded.
4. The UART receive interrupt is immediately restarted.

This method allows rapid detection of Bluetooth beacon signals without requiring full message parsing.

3.14.4 Timer Interrupt Handling

Timer interrupts play a crucial role in scheduling periodic tasks and controlling the alert system.

3.14.4.1 TIM3 Interrupt Handler

TIM3 generates an interrupt every one second. This interrupt is used to:

- Maintain system timing
- Control the buzzer ON/OFF pattern
- Track elapsed time for cooldown and alert cycles

Each interrupt increments a counter that determines whether the buzzer should be activated or deactivated.

3.14.4.2 PWM-Based Buzzer Control

The buzzer is driven using **TIM4 configured in PWM mode**. PWM is started and stopped inside the TIM3 interrupt handler based on predefined timing conditions. This separation of timing logic and signal generation ensures precise control over the alert pattern while minimizing CPU overhead.

3.14.5 UART Error Handling Mechanism

Communication errors are common in wireless and sensor-based systems, especially in noisy environments. To address this, the firmware implements a UART error callback mechanism.

When a UART error occurs:

1. The error callback function is triggered automatically.
2. The affected UART interface is identified.
3. UART reception is restarted immediately.
4. System operation continues without requiring a reset.

This automatic recovery mechanism prevents system crashes and ensures uninterrupted communication.

3.14.6 Buffer Overflow Protection

To prevent memory corruption, the algorithm includes buffer overflow checks during GPS data reception. If incoming GPS data exceeds the allocated buffer size:

- The buffer index is reset.
- Corrupted data is discarded.
- The system waits for the next valid sentence.

This protective measure enhances system stability and data integrity.

3.14.7 Interrupt Synchronization and Flag Management

Flags are used extensively to synchronize operations between interrupt handlers and the main execution loop. Interrupt handlers only set flags or update minimal variables, while the main loop processes the data. This design avoids lengthy operations inside interrupts and ensures deterministic system behavior.

3.14.8 Fail-Safe Behavior

In the event of severe errors or unexpected conditions:

- The system disables interrupts temporarily.
- The microcontroller enters a safe infinite loop.
- Visual indicators can be used for debugging.

This fail-safe approach prevents unpredictable behavior and protects hardware components.

3.14.9 Advantages of the Error Handling Strategy

The implemented error handling and interrupt management strategy provides several benefits:

- Ensures continuous system operation
- Prevents data loss during communication
- Minimizes CPU load
- Enhances real-time responsiveness
- Improves reliability in disaster environments

CHAPTER 4

RESULTS AND DISCUSSION

BLE-Based Human Presence Detection

The proposed system successfully detected human presence using Bluetooth Low Energy (BLE) beacon signals transmitted from a smartphone configured as an emergency beacon. During experimental evaluation, the rescue unit continuously scanned the surrounding area and identified BLE signals within the defined operational range. Detection occurred with minimal latency, enabling rapid identification of potential victims even during continuous patrol operations. The BLE scanning mechanism proved reliable under repeated trials, confirming stable performance without missed detections. These observations demonstrate that BLE technology is effective for proximity-based human detection, primarily due to its low power consumption, fast response time, and widespread availability in modern mobile devices.

GPS-Based Location Acquisition

Following successful BLE detection, the system accurately acquired real-time geographical location information using the GPS module. The receiver provided stable latitude and longitude coordinates after an initial fix period, and the acquired data remained consistent across multiple trials. The STM32 microcontroller correctly parsed the GPS data and synchronized it with the detected BLE signal information. This result confirms that GPS-based localization offers reliable and precise positioning, which is essential for guiding rescue teams to the exact victim location during flood rescue operations.

Priority Alert Generation and Formatting

Once location information was obtained, the system generated a priority rescue alert containing the detected victim's location details. The alert packet was structured to include essential parameters such as identification status and geographical coordinates, ensuring clarity and completeness of transmitted information. The formatted alert data was successfully prepared for wireless transmission without data corruption. This result

highlights the system's ability to automatically identify critical rescue locations and assign priority status, enabling faster and more informed decision-making at the rescue coordination center.

ZigBee-Based Long-Range Communication

The ZigBee wireless module successfully transmitted rescue alerts from the patrol unit to the remote rescue monitoring unit. Communication remained stable over extended distances, with minimal packet loss observed during testing. The system maintained consistent data transfer even during repeated alert transmissions. The robustness of ZigBee communication demonstrates its suitability for infrastructure-less disaster environments, where cellular networks are unreliable or completely unavailable.

Continuous Patrol and Multi-Detection Capability

The system demonstrated the ability to continue patrolling after detecting a victim while maintaining previously detected locations as priority zones. Multiple BLE beacon detections were successfully recorded during a single operational cycle without interrupting ongoing communication tasks. This capability ensures that rescue operations can scale efficiently in large flood-affected regions, allowing simultaneous identification of multiple victims and improved utilization of rescue resources.

Power Efficiency and Operational Stability

Power consumption analysis revealed that the system operates efficiently by activating the GPS module only when required and performing BLE scanning and ZigBee transmission intermittently. The embedded firmware optimized power usage while maintaining reliable system performance. The low-power design enables prolonged operation, which is critical during extended rescue missions where power availability is limited. The system remained operational and stable throughout the testing period without unexpected resets or failures.

Overall Results and System Performance Analysis

The experimental results collectively confirm that the Dual Mode Rescue Communicator performs reliably across all functional stages, including human detection, location acquisition, alert generation, and long-range communication. The integration of BLE and ZigBee technologies effectively combines fast short-range detection with reliable long-range data transmission, addressing key limitations of traditional rescue communication systems.

The system demonstrated low detection latency, accurate localization, stable communication, and efficient power usage, validating its suitability for disaster rescue applications. While environmental factors such as signal interference and terrain variations may influence real-world deployment, the obtained results clearly indicate that the proposed system provides a practical, scalable, and infrastructure-independent solution for flood rescue operations.

DISCUSSION

The obtained results validate the effectiveness of the **Dual Mode Rescue Communicator** in integrating short-range human detection, accurate localization, and reliable long-range communication into a unified rescue framework. The system addresses the limitations of manual and infrastructure-dependent rescue methods by providing automated detection and timely alert transmission. Although environmental factors such as signal interference and terrain variations may influence performance in real-world deployments, the experimental evaluation confirms the feasibility and practical applicability of the proposed system for flood disaster rescue operations.

CHAPTER 5

SUMMARY AND CONCLUSIONS

5.1. SUMMARY

Flood disasters often lead to severe disruption of communication infrastructure, delayed rescue response, and increased risk to human life. Traditional rescue mechanisms rely heavily on manual searching and infrastructure-dependent communication systems, which are ineffective under extreme disaster conditions. Recognizing these challenges, this mini project focused on the development of a **Dual Mode Rescue Communicator** aimed at improving rescue efficiency in flood-affected regions.

The proposed system integrates **Bluetooth Low Energy (BLE)** for short-range human presence detection and **ZigBee wireless communication** for long-range alert transmission. The use of BLE enables continuous scanning for human presence with minimal power consumption, while ZigBee ensures reliable communication over extended distances without relying on cellular networks. An **STM32 microcontroller** acts as the central controller, coordinating BLE scanning, GPS-based location acquisition, and ZigBee-based data transmission.

The system operates using a **patrol-based rescue approach**, where a mobile unit scans disaster-affected areas and automatically detects stranded victims. Once detection is achieved, **real-time GPS coordinates** are obtained and transmitted as priority alerts to a remote rescue coordination center. Experimental evaluation confirms that the system performs reliably in detecting victims, acquiring accurate location information, and transmitting alerts with minimal delay and low power consumption.

5.2. CONCLUSIONS

Based on the design, implementation, and experimental validation of the proposed system, the following conclusions are drawn:

- The **Dual Mode Rescue Communicator** successfully demonstrates automated human detection using BLE beacon signals.
- **GPS-based localization** provides accurate and reliable positioning information essential for effective rescue planning.
- **ZigBee wireless communication** ensures stable and long-range transmission of rescue alerts in infrastructure-deficient environments.
- The system achieves **low power operation**, making it suitable for prolonged rescue missions.
- Integration of sensing, localization, and communication into a single embedded platform enhances **response speed and operational efficiency.**
- The proposed approach effectively addresses the limitations of conventional rescue systems that depend on manual searching and cellular networks.
- The experimental results validate the **feasibility and practicality** of using dual-mode communication for disaster rescue applications.

Overall, the project confirms that combining short-range detection with long-range communication significantly improves the reliability and effectiveness of disaster rescue operations.

5.3. LIMITATIONS OF THE SYSTEM

Despite its successful performance, the proposed system has certain limitations that should be considered:

- The **BLE detection range** is limited and may be affected by environmental obstacles and signal interference.
- GPS accuracy may be reduced in **indoor environments or densely obstructed flood zones.**
- ZigBee communication range and reliability may vary depending on **terrain conditions and node placement.**

- The system is implemented and tested at a **prototype level**, and large-scale real-world deployment has not been evaluated.
- Autonomous navigation and obstacle avoidance are not included in the current system.
- Real-time integration with government disaster management platforms is outside the scope of this work.

These limitations highlight areas where further improvements and enhancements are possible.

5.4. FINAL REMARKS

This mini project successfully demonstrates the application of e**mbedded systems and wireless communication technologies** in disaster management and rescue operations. The proposed Dual Mode Rescue Communicator offers a **cost-effective, scalable, and infrastructure-independent solution** for flood rescue scenarios. By reducing reliance on manual operations and conventional communication networks, the system enhances rescue efficiency and safety.

The project bridges the gap between theoretical concepts and practical implementation, providing valuable insights into the design of intelligent rescue communication systems. The developed prototype serves as a strong foundation for future research and development in **smart disaster management technologies.**

5.5. RECOMMENDATIONS FOR THE FUTURE WORK

The proposed Dual Mode Rescue Communicator demonstrates the feasibility of integrating Bluetooth-based human detection, GPS-based localization, and ZigBee-based long-range communication for disaster rescue applications. While the current implementation relies on a microcontroller-based rescue unit, the system can be significantly enhanced by extending it into a **fully autonomous rescue platform**. One promising direction for future work is the design and development of an **autopilot-enabled rescue vehicle using a Raspberry Pi and ROS 2 framework**.

In the proposed future system, a **Raspberry Pi** can be employed as a high-level processing unit to handle autonomous navigation, decision-making, and sensor fusion. The Raspberry Pi can interface with the existing STM32-based rescue communicator, which continues to manage low-level communication tasks such as BLE detection, GPS data acquisition, and ZigBee transmission. This division of responsibilities allows efficient utilization of both microcontroller and single-board computer capabilities.

Autonomous navigation can be achieved using the **Robot Operating System (ROS 2)**, which provides a modular, scalable, and real-time communication framework suitable for robotic applications. ROS 2 nodes can be developed to process sensor data, generate navigation commands, and monitor system status. The use of ROS 2 enables distributed processing, reliable message passing, and easy integration of additional sensors and algorithms.

A **gyroscopic sensor (IMU)** can be integrated into the autopilot system to provide orientation, angular velocity, and motion data. By combining gyroscope data with GPS information, the rescue vehicle can maintain stable movement, correct its heading, and navigate accurately in challenging flood environments. Sensor fusion techniques can be applied to improve positional accuracy and robustness under dynamic conditions.

In a flood rescue scenario, the autonomous system can be implemented on a **self-navigating rescue boat**. Upon receiving victim location data from the Dual Mode Rescue Communicator, the ROS 2-based autopilot can automatically plan a path, avoid obstacles, and navigate toward the detected location without continuous human intervention. This autonomous operation can significantly **reduce rescue response time**, minimize human risk, and enable simultaneous rescue missions in multiple locations.

Further enhancements may include the integration of cameras for visual feedback, obstacle detection sensors, and machine learning algorithms for intelligent decision-making. Overall, the development of an autonomous autopilot system using Raspberry Pi, ROS 2, and gyroscopic sensors represents a scalable and impactful extension of the proposed project, with the potential to greatly improve the effectiveness of disaster rescue operations.

- Integration with an **autonomous rescue boat equipped with BeaconLink** for automated patrol and victim assistance.
- Use of **mesh networking** with multiple ZigBee nodes to extend communication coverage.
- Incorporation of **additional sensors** such as ultrasonic, IMU, or camera modules for obstacle detection.
- Implementation of **real-time monitoring and data visualization** at the rescue control center.
- Adoption of **machine learning algorithms** for intelligent victim detection and prioritization.
- Enhancement of power systems using **solar energy or energy harvesting techniques.**
- Development of a **mobile application** for rescue coordination and alert management.

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